

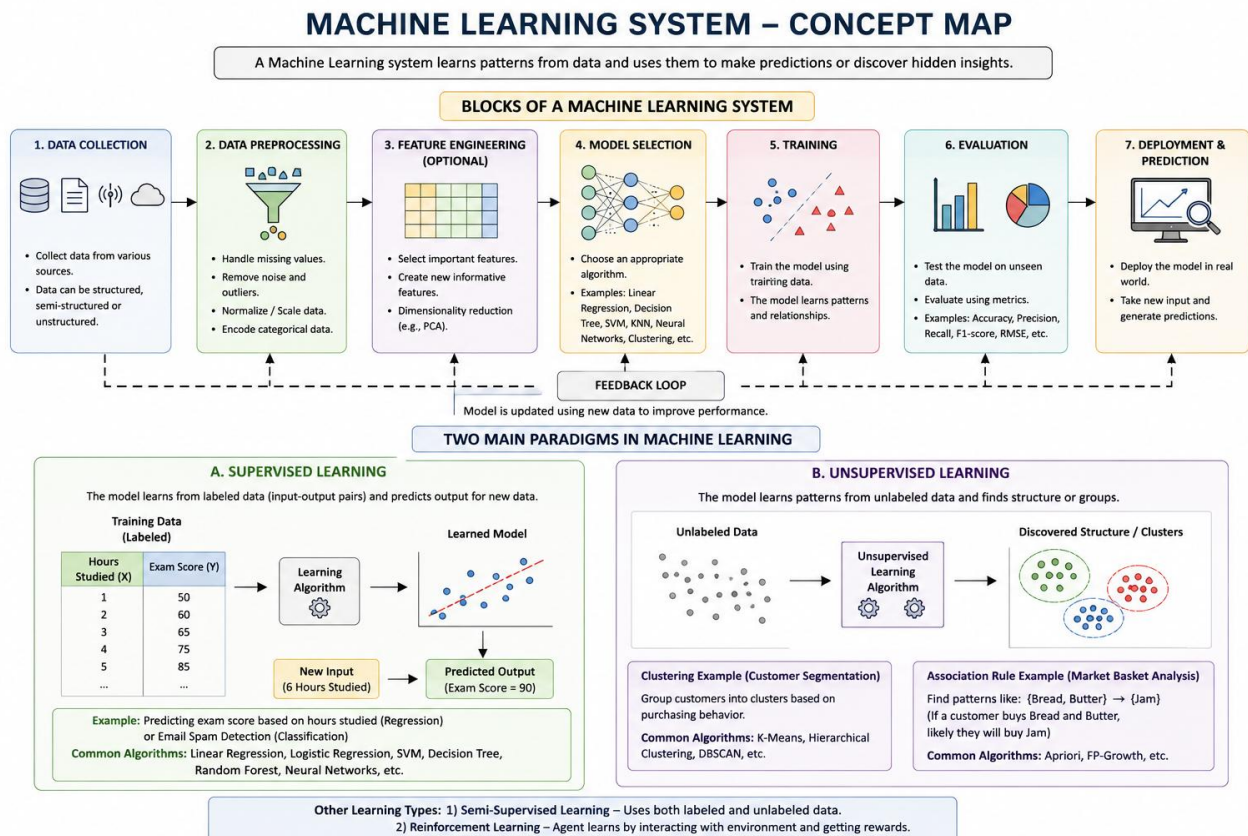
CO1-AT2

Chart Construction Test

Name: Rishwanth. L
Reg No: 192424100

1. Machine Learning System

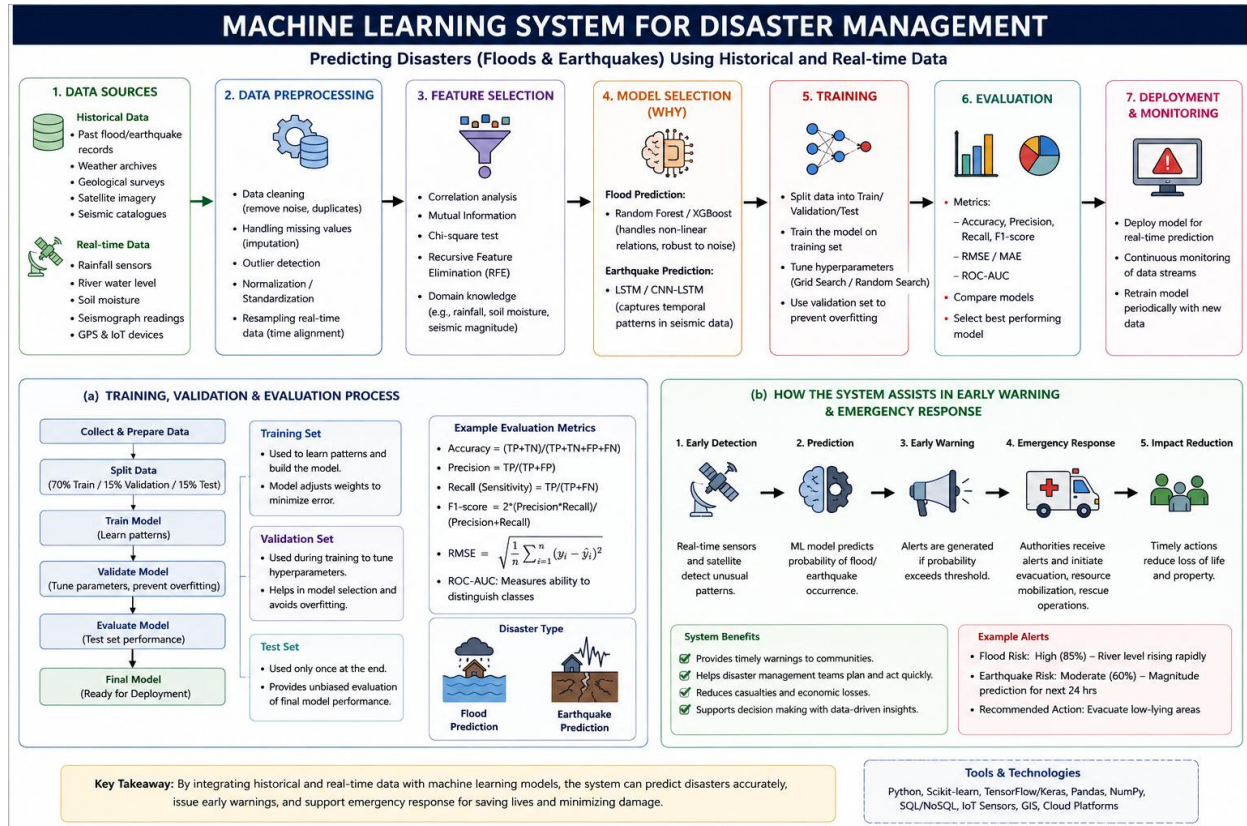
With the necessary sketch, explain all the possible blocks in the machine-learning system. Give example for showing the concept of the supervised and Un-supervised learning methods in the machine learning scheme.



2. Disaster Management System

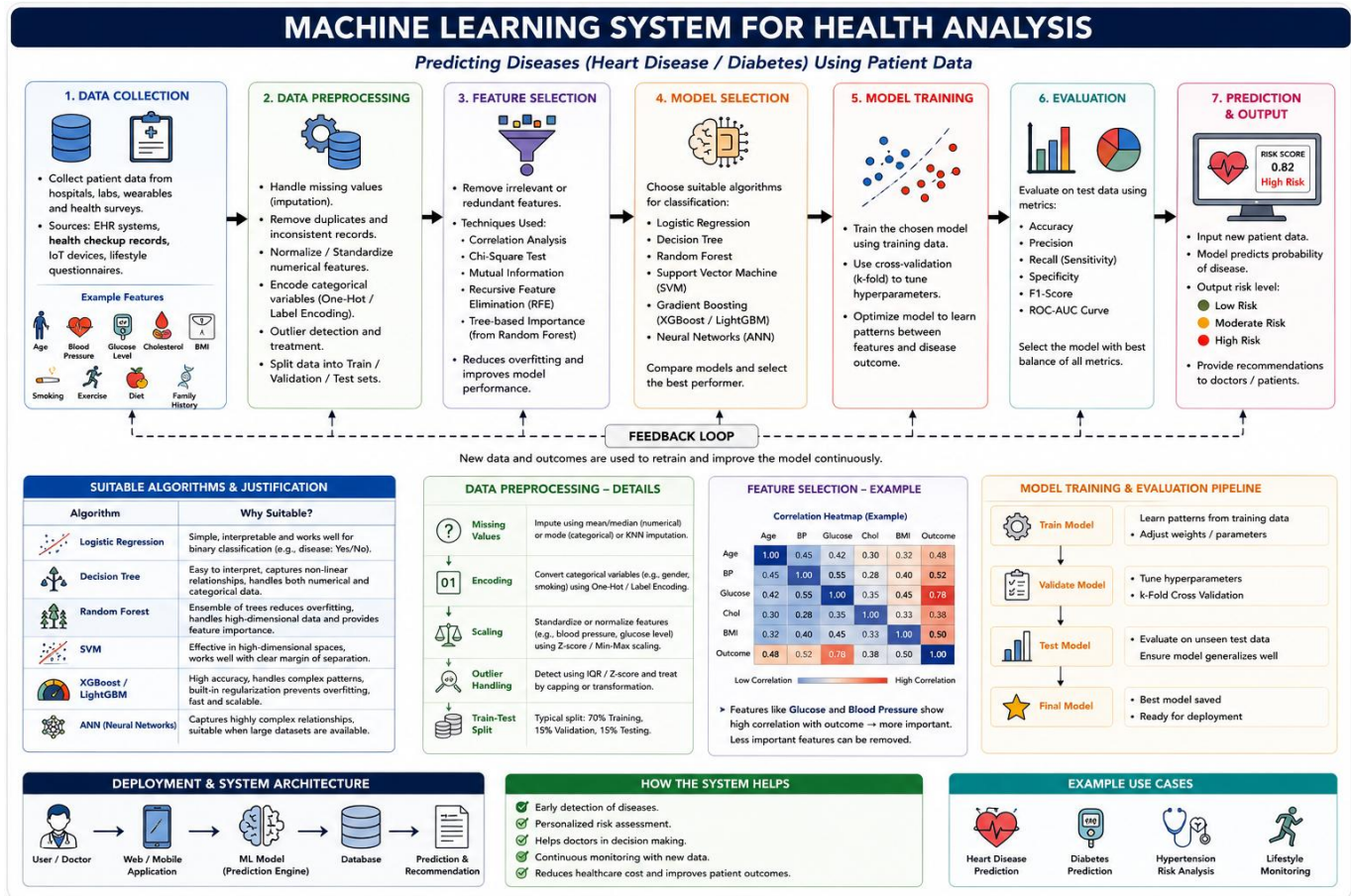
Design a machine learning system for disaster management to predict events such as floods and earthquakes using historical and real-time data. Explain the choice of model, data preprocessing, and feature selection techniques.

- Describe the steps involved in training, validating, and evaluating the model.
- Explain how the system can assist in early warning and emergency response.



3. Health Analysis

Design a machine learning system for health analysis to predict diseases such as heart disease or diabetes using patient data including age, blood pressure, glucose level, and lifestyle factors. Explain the choice of suitable algorithms and justify their selection. Describe the steps involved in data preprocessing, feature selection, and model training to ensure accurate and reliable predictions.



4. Typical Machine Learning Pipeline

Explain the architecture of a typical machine learning pipeline with a neat diagram. Discuss the role of each stage including data collection, preprocessing, feature engineering, model training, evaluation, and deployment. Also, differentiate between training error and generalization error with suitable examples.

